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Data Analytics Mini Project

LINK: [HTTPS://WWW.KAGGLE.COM/DATASETS/AMINEBANNOUR/MESSYSALESDATA](https://www.kaggle.com/datasets/aminebannour/messysalesdata)

Overview of the Uncleaned Dataset

The raw sales dataset contains transaction records across multiple countries, products, and sales representatives. However, a closer look reveals several consistency and quality issues that need to be addressed before the data can be used for any meaningful analysis. These issues, if left uncleaned, would lead to inaccurate groupings, wrong totals, and unreliable reporting.

The screenshot below shows a sample of the raw, uncleaned data as it was received:

invoice_id	order_date	customer_name	customer_email	country	city	product_name	category	quantity	unit_price	discount	payment_method	sales_rep	region	shipping_cost
INV-1000	02-03-2015	Dr. Jeffery Bailey	bademail.com	Italy	Heatherton	Office Chair	Furniture	2	579.16	15	Cash		Europe	15
INV-1000	18-09-2004	Sarah Rivas		Tunisia	Lake Andrew	Keyboard Mechanical	Electronics	0	1168.73	5	PayPal		Europe	10
INV-1000		Robert Orozco	bademail.com	Italy	Jonesport	Office Chair	Furniture	4	166.24		Cash		Europe	10
INV-1000	22-02-1975	Kimberly Williams		tunisia	Griffithfurt	Keyboard Mechanical	Electronics	5	1035.19	150	Cash		Africa	15
INV-1000	16-12-1977	Mr. Arthur Ford	bademail.com	tunisia	New James	Wireless Mouse	electronics	3	491.85	15	credit card	Monica	Europe	15
INV-1000	21-09-1977	Carolyn Wright	meganrobles@example.com	tunisia	West Jason	Standing Desk	furniture	1	948.25	15	paypal	Michael	Europe	15
INV-1000		Kimberly Craig	zrivera@example.net	tunisia	Allenhaven	Tablet Air	Electronics	0	1331.75	10	paypal		Africa	
INV-1000	15-10-1993	Kristy Wilson	nancy25@example.com	Spain	Port John	Printer HP 2024	electronics	2	167.13	15	credit card	James	Africa	10
INV-1000	07-11-1986	Nathan Nichols		Italy	Jamesside	Office Chair	Furniture	-1	216.03	15	paypal		North America	5
INV-1000	24-08-2009	Omar Roberts		Italy	Port Samuel	Monitor 24 inch	Electronics	2	1278.03	0	credit card	Valerie	North America	
INV-1000	02-12-2014	Catherine Anderson	bademail.com	Germany	Mcgrathborough	Notebook A4	Stationery	5	1357.55	150	Credit Card	Lauren	North America	10
INV-1000	02-03-2011	Ashlee Delacruz		Italy	New Evan	Office Chair	Furniture	1	321.99	0	Cash		Africa	15
INV-1000	09-03-2001	Tammy Harrell	bademail.com	Italy	Port Jasmine	Monitor 24 inch	Electronics	3	452.9	150	credit card	Gail	Africa	
INV-1000	19-07-1995	William Knight	leedwin@example.com	U.S.A	Lake Joshuaburgh	USB-C Cable	Accessories	3	300.39	0	Bank Transfer	Michael	North America	
INV-1000	24-10-1998	Andrea Murphy	brennansharon@example.org	Italy	Shepherdstown	Office Chair	Furniture	0	59.42	10	PayPal		North America	10
INV-1000	04-08-2003	Priscilla Shields	stephen73@example.org	U.S.A	East Robert	Wireless Mouse	electronics	5	225.95	0	Credit Card		Europe	
INV-1000		Lindsey Baker	marilyn95@example.com	Tunisia	Gonzalezberg	Standing Desk	furniture	5	1316.1		Credit Card	Jose	North America	5
INV-1000	16-06-2004	Aaron Cross	bademail.com	USA	Katherineland	USB-C Cable	Accessories	3	835.59	5	credit card		Europe	
INV-1001	09-08-2011	Mckenzie Thomas		Italy	Dorseychester	Laptop Pro 15	Electronics	1	1200.35	10	Cash		Africa	5
INV-1001	04-04-1984	Rachel Boone	bademail.com	tunisia	New Dan	Office Chair	Furniture	5	79.47	0	Credit Card	Brett	Europe	10
INV-1001	08-10-1982	Tina Johnson		tunisia	Crystalton	Printer HP 2024	electronics	2	447.38	5	credit card	Ricky	Africa	
INV-1001		James Castro	bademail.com	Germany	Jenniferside	Keyboard Mechanical	Electronics	1	789.44	15	PayPal	Joy	North America	5
INV-1001	11-06-1988	Levi Silva		Italy	Jerrystad	Laptop Pro 15	Electronics	0	18.46	150	credit card		Africa	10
INV-1001	11-08-2013	Lisa Moses	bademail.com	USA	North Douglasville	Printer HP 2024	electronics	2	156.75	5	credit card	Tonya	North America	5
INV-1001	25-09-1976	Alyssa Cox PhD		Germany	Victoriafurt	Tablet Air	Electronics	-1	1347.32		credit card	Mark	Europe	15
INV-1001	30-10-1996	David Morris		U.S.A	Tonitown	Tablet Air	Electronics	5	577.42	150	PayPal		Europe	15
INV-1001	16-01-1977	Kimberly Allen	bademail.com	France	South Nancyburgh	USB-C Cable	Accessories	3	342.39	0	paypal	Carrie	North America	5

Data Cleaning Procedures

We The following steps were performed to clean and standardize the dataset:

1. **Standardized Country Names:** Corrected inconsistent country name entries (e.g., "tunisia" → "Tunisia", "U.S.A" → "U.S.A") to ensure uniform formatting.
2. **Fixed Invalid Email Addresses:** Replaced placeholder or invalid emails (e.g., "bademail.com") with "no.email.com" and filled missing email fields accordingly.
3. **Handled Missing Dates:** Replaced blank Date entries with "Average of the Dates" to maintain data completeness.
4. **Extracted Year from Order Date:** Added a new "Year" column extracted from the Order Date field to enable time-based analysis.
5. **Standardized Category & Payment Method Columns:** Applied consistent title-case formatting across the Category and Payment Method columns to eliminate mixed-case entries.
6. **Rounded Numeric Fields:** Rounded unit price and other numeric values to whole numbers to improve readability.
7. **Handled Missing Discount Amount & Shipping Cost Values:** Replaced blank Discount & Shipping Cost column with "Mode" Function For the calculations.
8. **Handled Missing Sales Rep Values:** Replaced blank Sales Rep entries with "Unknown" to maintain data completeness.

Cleaned Dataset

After applying all the cleaning steps above, the dataset is now consistent, complete, and ready for analysis. The screenshot below shows the resulting cleaned dataset:

Invoice Id	Order Date	Year	Customer Name	Email	Country	City	Product Name	Category	Quantity	Amount	Discount Amount	Payment Method	Sales Rep	Region	Actual Shipping Cost
INV-1000	02-03-2015	2015	Dr. Jeffery Bailey	bademail.com	Italy	Heatherton	Office Chair	Furniture	2	579	15	Cash	Unknown	Europe	15
INV-1000	18-09-2004	2004	Sarah Rivas	no.email.com	Tunisia	Lake Andrew	Keyboard Mechan	Electronics	0	1169	5	Paypal	Unknown	Europe	10
INV-1000	29-07-1997	1997	Robert Orozco	bademail.com	Italy	Jonesport	Office Chair	Furniture	4	166	5	Cash	Unknown	Europe	10
INV-1000	22-02-1975	1975	Kimberly Williams	no.email.com	Tunisia	Griffthfurt	Keyboard Mechan	Electronics	5	1035	150	Cash	Unknown	Africa	15
INV-1000	16-12-1977	1977	Mr. Arthur Ford	bademail.com	Tunisia	New James	Wireless Mouse	Electronics	3	492	15	Credit Card	Monica	Europe	15
INV-1000	21-09-1977	1977	Carolyn Wright	meganrobles@example.com	Tunisia	West Jason	Standing Desk	Furniture	1	948	15	Paypal	Michael	Europe	15
INV-1000	29-07-1997	1997	Kimberly Craig	zrivera@example.net	Tunisia	Allenhaven	Tablet Air	Electronics	0	1332	10	Paypal	Unknown	Africa	5
INV-1000	15-10-1993	1993	Kristy Wilson	nancy25@example.com	Spain	Port John	Printer HP 2024	Electronics	2	167	15	Credit Card	James	Africa	10
INV-1000	07-11-1986	1986	Nathan Nichols	no.email.com	Italy	Jamesside	Office Chair	Furniture	-1	216	15	Paypal	Unknown	North America	5
INV-1000	24-08-2009	2009	Omar Roberts	no.email.com	Italy	Port Samuel	Monitor 24 inch	Electronics	2	1278	5	Credit Card	Valerie	North America	5
INV-1000	02-12-2014	2014	Catherine Anderson	bademail.com	Germany	Mcgrathborough	Notebook A4	Stationery	5	1358	150	Credit Card	Lauren	North America	10
INV-1000	02-03-2011	2011	Ashlee Delacruz	no.email.com	Italy	New Evan	Office Chair	Furniture	1	322	5	Cash	Unknown	Africa	15
INV-1000	09-03-2001	2001	Tammy Harrell	bademail.com	Italy	Port Jasmine	Monitor 24 inch	Electronics	3	453	150	Credit Card	Gail	Africa	5
INV-1000	19-07-1995	1995	William Knight	leedwin@example.com	U.S.A	Lake Joshuaburgh	USB-C Cable	Accessories	3	300	5	Bank Transfer	Michael	North America	5
INV-1000	24-10-1998	1998	Andrea Murphy	brennansharon@example.org	Italy	Shepherdstown	Office Chair	Furniture	0	59	10	Paypal	Unknown	North America	10
INV-1000	04-08-2003	2003	Priscilla Shields	stephen73@example.org	U.S.A	East Robert	Wireless Mouse	Electronics	5	226	5	Credit Card	Unknown	Europe	5
INV-1000	29-07-1997	1997	Lindsey Baker	marilyn95@example.com	Tunisia	Gonzalezberg	Standing Desk	Furniture	5	1316	5	Credit Card	Jose	North America	5
INV-1000	16-06-2004	2004	Aaron Cross	bademail.com	U.S.A	Katherineland	USB-C Cable	Accessories	3	836	5	Credit Card	Unknown	Europe	5
INV-1001	09-08-2011	2011	Mckenzie Thomas	no.email.com	Italy	Dorseychester	Laptop Pro 15	Electronics	1	1200	10	Cash	Unknown	Africa	5
INV-1001	04-04-1984	1984	Rachel Boone	bademail.com	Tunisia	New Dan	Office Chair	Furniture	5	79	5	Credit Card	Brett	Europe	10
INV-1001	08-10-1982	1982	Tina Johnson	no.email.com	Tunisia	Crystaltown	Printer HP 2024	Electronics	2	447	5	Credit Card	Ricky	Africa	5
INV-1001	29-07-1997	1997	James Castro	bademail.com	Germany	Jenniferside	Keyboard Mechan	Electronics	1	789	15	Paypal	Joy	North America	5
INV-1001	11-06-1988	1988	Levi Silva	no.email.com	Italy	Jerrystad	Laptop Pro 15	Electronics	0	18	150	Credit Card	Unknown	Africa	10
INV-1001	11-08-2013	2013	Lisa Moses	bademail.com	U.S.A	North Douglasville	Printer HP 2024	Electronics	2	157	5	Credit Card	Tonya	North America	5
INV-1001	25-09-1976	1976	Alyssa Cox PhD	no.email.com	Germany	Victoriafurt	Tablet Air	Electronics	-1	1347	5	Credit Card	Mark	Europe	15
INV-1001	30-10-1996	1996	David Morris	no.email.com	U.S.A	Tonitown	Tablet Air	Electronics	5	577	150	Paypal	Unknown	Europe	15
INV-1001	16-01-1977	1977	Kimberly Allen	bademail.com	France	South Nancyburgh	USB-C Cable	Accessories	3	342	5	Paypal	Carrie	North America	5
INV-1001	21-02-1980	1980	Timothy Dean	no.email.com	Spain	South Omarland	Notebook A4	Stationery	1	749	5	Credit Card	Unknown	Europe	5
INV-1001	12-08-2014	2014	Michelle Taylor	bademail.com	U.S.A	East Thomasland	Tablet Air	Electronics	2	22	5	Bank Transfer	Unknown	Africa	5
INV-1001	10-06-2024	2024	Tonya Freeman	matthew33@example.org	Tunisia	Jordanmouth	Standing Desk	Furniture	-1	235	5	Bank Transfer	Christopher	Africa	5

DAX Measures & Calculated Columns In Power BI

The following DAX measures and calculated fields power every visual in the dashboard:

Core Sales Measures

Calculated Column: IF condition like, When Quantity is Equal to “0” then “No sale”, “<0” then “Return”, “>0” the it should be “Sale”.

Total Gross Sale: SUM of the Amount column , calculate only the amount is equal to the “Sale” in the Sales Status Column this is the condition in this dataset to calculate.

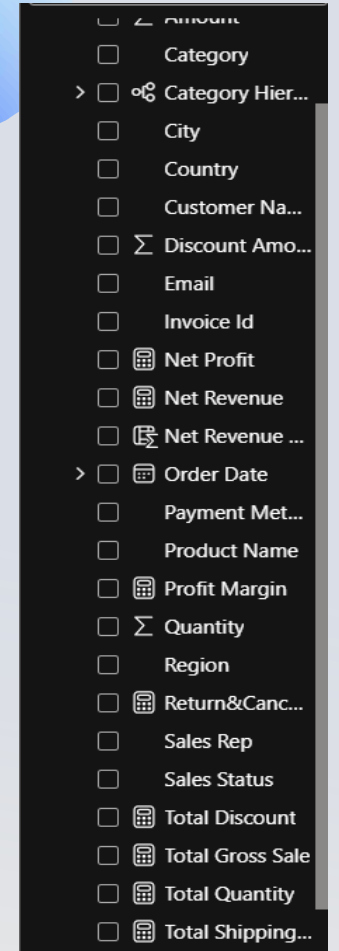
Total Discount: SUM of the Discount Amount column, calculate only the Discount is equal to the “Sale” in the Sales Status Column.

Total Shipping Costs: SUM of the Actual Shipping Cost column, calculate only the Shipping is equal to the “Sale” in the Sales Status Column, total logistics expense across all orders.

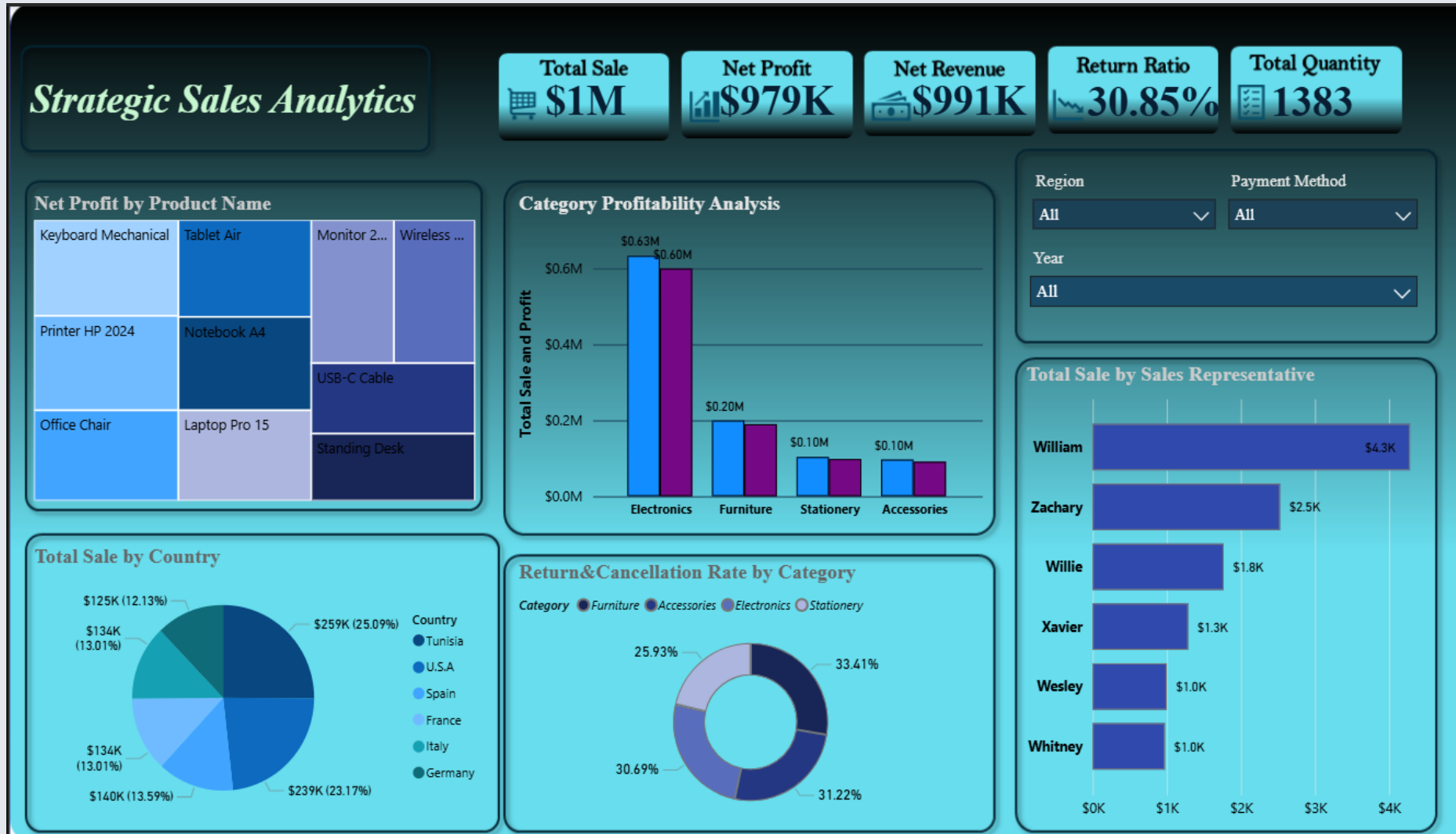
Net Revenue: Calculated as Total Gross Sale minus Total Discount. Reflects adjusted revenue after price reductions.

Net Profit: Net Revenue minus Total Shipping Costs. The final bottom-line profitability measure.

The DAX measures transform a cleaned flat Excel table into a dynamic, interactive, and analytically powerful business dashboard. Without them, the data would simply be a grid of rows - with them, it becomes a decision-making tool.



Data Visualization By Using Power BI



Dashboard

1. Category Profitability

Electronics is the undisputed superstar of the business, bringing in \$0.63M. Even better, almost all of that revenue turns into pure profit, showing excellent profit margins. Furniture, Stationery, and Accessories trail far behind.

2. Top Profitable Products

The "Keyboard Mechanical" and "Tablet Air" are the main money-makers keeping the business running. Items like laptops and monitors do okay, while small accessories like cables and standing desks bring in very little profit.

3. Where the Sales Come From

Sales are spread across different countries, but Tunisia takes the #1 spot, bringing in a quarter of all sales (\$259K). Spain is a strong second place (\$239K), while the rest of the sales are split evenly among the U.S.A. and Europe

4. Returns & Cancellations

Stationery has the worst return rate at 33.41%, closely followed by Electronics at 31.22%. Because Electronics is the top seller, having nearly a third of its orders returned or canceled means the business is losing a massive amount of money.

5. Sales Team Performance

William is the absolute MVP of the sales team, bringing in a whopping \$4.3K. Zachary does a great job in second place at \$2.5K. The rest of the team lags behind, with the lowest-performing members stuck around the \$1.0K mark

Future Predictions Report

- **Diversification** : To secure future revenue growth, the business must expand sales in Furniture and Accessories to reduce its risky over-reliance on the Electronics category.
- **Optimization** : Implementing quality audits and better packaging can reduce high return rates to 18–22%, reclaiming up to \$150K in lost annual revenue.
- **Expansion**: Strategic marketing and inventory budgets should be funneled directly into Tunisia and Spain to capitalize on these high-performing, high-return markets.
- **Upskilling**: Providing structured coaching to low- and mid-tier representatives will help close the massive internal performance gap and drive significant incremental sales growth.

KEY RECOMMENDATIONS

- **Increase Electronics Stock** - Expand inventory & run targeted campaigns
- **Audit Furniture & Accessories** - Reduce returns via quality checks
- **Focus on Top 3 Markets** - Tunisia, U.S.A & Spain priority
- **Train Sales Team** - Replicate William's winning method

Thank you

